

Managing System-Wide Login Scripts



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Overview



Central Login Scripts / Skeleton Directory

- Investigating the user home directory
- Changing the skeleton content
- Login and logout scripts
- Implementing centralized login scripts
 - `/etc/bash.bashrc`
 - `/etc/profile`
 - `/etc/profile.d`



```
$ sudo useradd -m test_user  
$ ls -A ~test_user  
$ ls -A /etc/skel
```

Default User Content

The user home directory is created from **/etc/skel** by default. The content only affects new home directories and not existing content.

Default Content from /etc/skel

`.profile`

Executes once per login for bash and bourne shells

`.bashrc`

Executes for interactive bash shells each time bash executes

`.bash_logout`

Executes when a login shell is exited



Demo



Login Scripts and User Templating

- New user with `useradd`
- List home directory
- Compare with `/etc/skel`
- Add content to `/etc/skel`



Login Shells

.profile

Executed at the start of a user session

.bash_logout

Executed when users exits from login shell



```
$ umask  
$ grep '#umask' .profile  
$ sed -i '/^#umask/s/^#/' .profile ##uncomment the umask setting  
$ sed -i '/^umask/s/^/#/' .profile ##to comment the line
```

Configuring an Individual Umask

The default umask in Ubuntu is 0002 for standard users, if you want to configure this differently it can be configured in your login shell. We can easily comment and uncomment lines using sed.

```
$ cat .bash_logout  
$ /usr/bin/clear_console  
$ sed -i '$a clear' .bash_logout #appends to the last line  
$ sed -i '$d' .bash_logout #deletes the last line
```

Clearing Screen on Logout

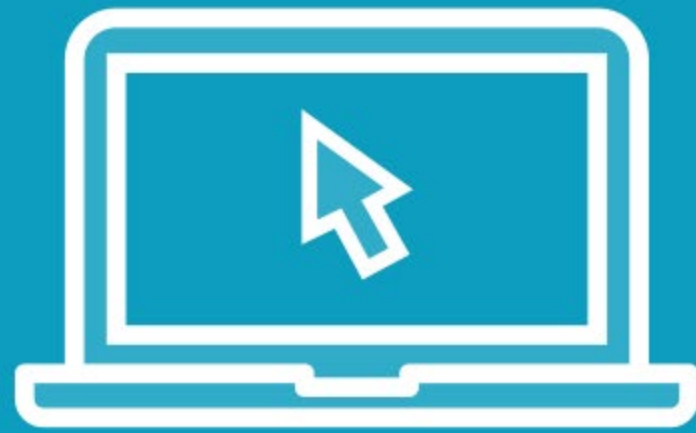
The default logout script will clear the screen for physical consoles but not pseudo-consoles. We can add in the clear command to add privacy upon logout. Again, using sed we can append the the end of the file and delete the last line. The dollar symbol represents the last line in both examples.


```
$ sudo apt update && sudo apt install -y bash-doc  
$ ls /usr/share/doc/bash/examples/startup-files/
```

Login Script Examples

Installing the package bash-doc will add example login scripts

Demo



Working with Login Shells:

- bash-doc
- .profile
- .bash_logout

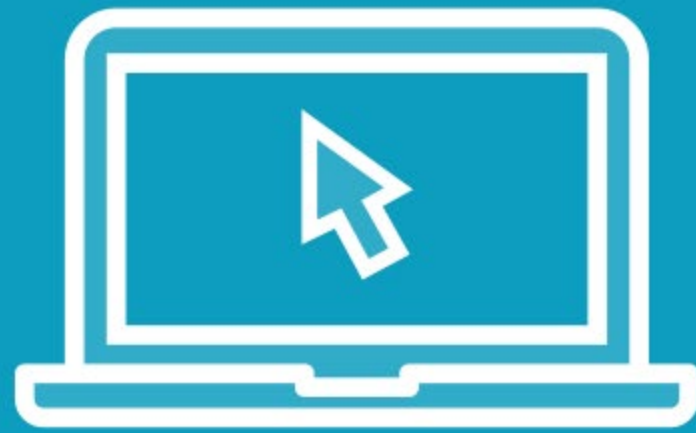


```
$ ls /etc/profile
$ ls /etc/profile.d
$ ls /etc/bash.bashrc
$ echo "umask 027" | sudo tee /etc/profile.d/umask.sh
```

Common Settings

Command setting for all users can be configured with `/etc/profile` or `/etc/bash.bashrc` . As the profile only runs once, upon login, this is preferred and has its own extension directory, `/etc/profile.d/` where you can add `.sh` files.

Demo



Adding Common Settings:

- umask
- aliases
- shell options



Summary



Login Scripts

- /etc/skel
- .profile
- .bashrc
- .bash_logout
- /etc/profile
- /etc/profile.d
- /etc/bash.bashrc



Managing Pluggable Authentication Modules (PAM)

